

Studying the relationship between two sets of data

Scatter diagrams, correlation, the line of best fit — and the one warning that must accompany all of it.

LESSON 71–73

3 × 40 MIN

ALGEBRA 11, §4.5

IGCSE E16.1

LESSON CLOCK

18'

26'

30'

28'

30'

3

Scatter diagrams	18 min
Describing correlation	26 min
The line of best fit	30 min
Correlation is not causation	28 min
Practice	30 min
Homework	3 min

LEARNING OBJECTIVES

- Plot and read a scatter diagram.
- Describe correlation by type and strength.
- Draw a line of best fit through the mean point and use it to predict.
- Distinguish correlation from causation, and interpolation from extrapolation.

TEXTBOOK

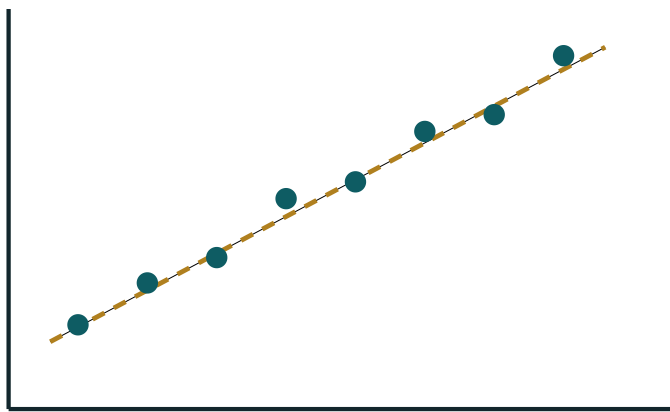
National	pp. 339–356
Cambridge	Core & Extended, pp. 413–424
Workbook	IGCSE Exercise 16.1

01

Explanation

Scatter diagrams

Each individual contributes **one point** (x, y) — the two measurements made on the same person or object. The picture then shows whether the two vary together.



positive correlation

Each point is one individual, measured twice. The pattern, not the points, is the message.

CHOOSE THE AXES DELIBERATELY

If one variable might explain the other, put the explanatory variable on the horizontal axis. “Hours revised” goes on x ; “mark” goes on y . It changes nothing mathematically and everything about readability.

Describing correlation

PATTERN	NAME	EXAMPLE
rising, tight	strong positive	height and arm span
rising, loose	weak positive	revision hours and mark
falling, tight	strong negative	age of a car and its price
no pattern	no correlation	shoe size and mathematics mark
a curve	a relationship, but not linear	height of a thrown ball against time

ALWAYS GIVE BOTH WORDS

“Correlation” alone is not an answer. State the **type** — positive or negative — and the **strength** — strong, moderate or weak. Two words, two marks.

The line of best fit

TWO RULES FOR DRAWING IT BY EYE

1 It must pass through the **mean point** $(\bar{x}, \bar{y})$. **2** It should have roughly as many points above it as below, and follow the trend.

Once drawn, it predicts: read up from an x to the line and across to a y . Its gradient is the rate of change — “each extra hour of revision is worth about 4 marks”.

PREDICTION	NAME	RELIABLE?
inside the range of the data	interpolation	yes, if the correlation is strong
outside the range	extrapolation	no — the pattern may not continue

Predicting a mark for 3 hours from data covering 1 to 8 hours is interpolation.
Predicting for 40 hours is extrapolation, and the line would eventually promise a mark above 100.

Correlation is not causation

THE ONE WARNING THAT MUST ALWAYS ACCOMPANY A CORRELATION

Two variables can move together for three different reasons: one causes the other; both are caused by a third; or it is coincidence. A scatter diagram cannot tell which.

CORRELATED	REAL EXPLANATION
ice-cream sales and drownings	both rise with hot weather
shoe size and reading age in children	both rise with age
number of firefighters and damage	both rise with the size of the fire

Establishing causation needs an **experiment** — changing one variable deliberately and holding others fixed — not a larger data set.

02 Worked examples

EXAMPLE 1 Describe the correlation between the age of a car and its price.

- ① As age rises, price falls.
Negative.
- ② The points lie fairly close to a line.
Strong.

ANSWER

Strong negative correlation

EXAMPLE 2 Data on revision hours 1–8 gives a best-fit line $y = 42 + 4.5x$. Predict the mark for 5 hours, and for 30.

- ① $42 + 22.5 = 64.5$
Interpolation — inside the range.
- ② $42 + 135 = 177$
Impossible.
- ③ The second is extrapolation far outside the data.

ANSWER

≈ 65 ; the second prediction is invalid

EXAMPLE 3 Ice-cream sales and drownings are strongly correlated. Does ice cream cause drowning?

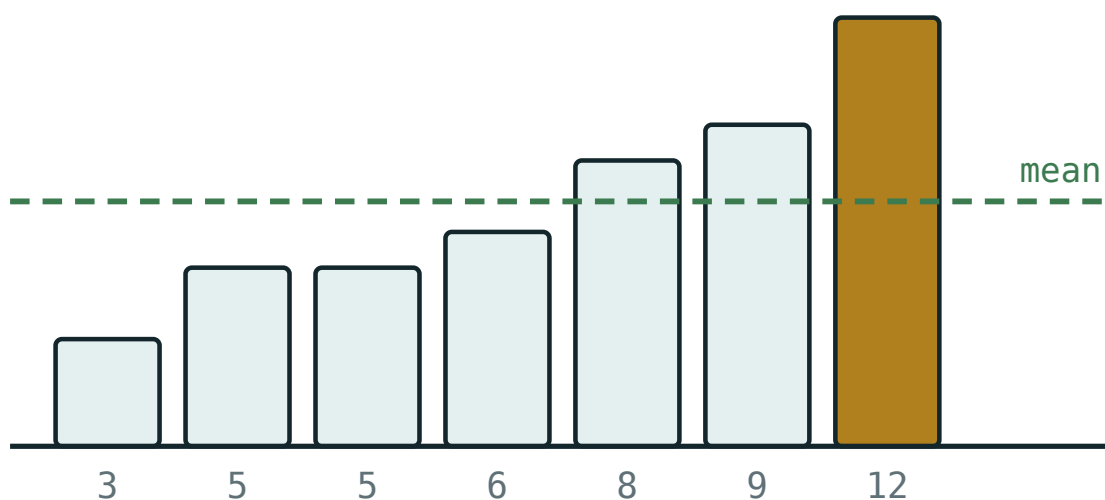
- ① Both rise in hot weather.
- ② A third variable explains both.
- ③ Correlation is not causation.

ANSWER

No — temperature is the common cause

03 Interactive model

Plot the class's height against arm span and draw the line through the mean point.

A scatter and its line

values 3, 5, 5, 6, 8, 9, 12

mean 6.86

median 6

mode 5

range 9

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Bivariate data	Ikki o'zgaruvchili ma'lumot	Двумерные данные
Scatter diagram	Korrelatsion diagramma	Диаграмма рассеяния
Correlation	Korrelyatsiya	Корреляция
Positive correlation	Musbat korrelyatsiya	Положительная корреляция
Negative correlation	Manfiy korrelyatsiya	Отрицательная корреляция
Line of best fit	Eng mos chiziq	Линия наилучшего соответствия
Mean point	O'rta nuqta	Средняя точка
Interpolation	Interpolyatsiya	Интерполяция
Extrapolation	Ekstrapolyatsiya	Экстраполяция
Causation	Sababiy bog'lanish	Причинная связь

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Age of a car against price shows:

strong positive

strong negative

no correlation

a curve

The line of best fit must pass through:

the origin

the mean point

the first point

the highest point

Predicting outside the data range is:

interpolation

extrapolation

correlation

causation

Correlation implies causation:

always

never

sometimes, but it cannot be told from the data

only if strong

Describing a correlation needs:

one word

type and strength

a number only

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Height and arm span

ANSWER strong positive

2. Age of a car and its price

ANSWER strong negative

3. Shoe size and mathematics mark

ANSWER no correlation

4. Temperature and ice-cream sales

ANSWER positive

5. The line of best fit passes through

ANSWER the mean point

6. Predicting inside the range is called

ANSWER interpolation

7. Predicting outside is called

ANSWER extrapolation

Medium · the standard question

7 PROBLEMS

1. $y = 42 + 4.5x$; predict for $x = 5$

ANSWER 64.5

2. Why is the prediction for $x = 30$ invalid?

ANSWER Far outside the data; it exceeds 100

3. Interpret the gradient 4.5

ANSWER about 4.5 marks per extra hour

4. Interpret the intercept 42

ANSWER the predicted mark with no revision

5. Mean point of $(1,3),(2,5),(3,7)$

ANSWER $(2, 5)$

6. Does a strong correlation prove causation?

ANSWER no

7. Firefighters and damage: the real cause

ANSWER the size of the fire

Hard · stretch and reason

7 PROBLEMS

1. Points $(1,4), (2,7), (3,9), (4,12), (5,14)$: the mean point

ANSWER $(3, 9.2)$

2. Estimate the gradient of the best-fit line for those points

ANSWER ≈ 2.5

3. Hence estimate y at $x = 3.5$

ANSWER ≈ 10.5

4. Why must the line pass through the mean point?

ANSWER The least-squares line always does

5. Give a pair of variables with a strong correlation and no causal link

ANSWER e.g. stork numbers and birth rates

6. A curve, not a line, fits the data. What does that show?

ANSWER A relationship, but not a linear one

7. Design an experiment to test whether revision causes better marks

ANSWER Assign revision time at random and compare groups

07 Homework

Homework — 5 tasks

Every correlation described with two words; every prediction labelled interpolation or extrapolation.

- Plot the eight points $(1,5), (2,6), (3,9), (4,10), (5,12), (6,14), (7,15), (8,18)$ and describe the correlation.
- Find the mean point and draw the line of best fit through it.
- Use your line to predict y at $x = 4.5$ and at $x = 20$, saying which prediction is reliable.
- Give two pairs of variables that are correlated without one causing the other, with the real explanation.

5. Explain in three sentences the difference between interpolation and extrapolation.